

HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 12 Higher School Certificate Trial Examination
Term 3 2024

STUDENT NUMBER: _____

**General
Instructions:**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- A NESA reference sheet is provided

**Total Marks:
100**

Section I – 10 marks (pages 3–6)

- Attempt Questions 1– 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8 - 29)

- Attempt Questions 11 - 34
- Allow about 2 hours and 45 minutes for this section

Q1 - 10	Q11 – 34	Total
/10	/90	/100

Outcomes assessed: MA 12 – 1, 3, 4, 5, 6, 7, 8, 10

This assessment task constitutes 30% of the Higher School Certificate Course School Assessment

Section I

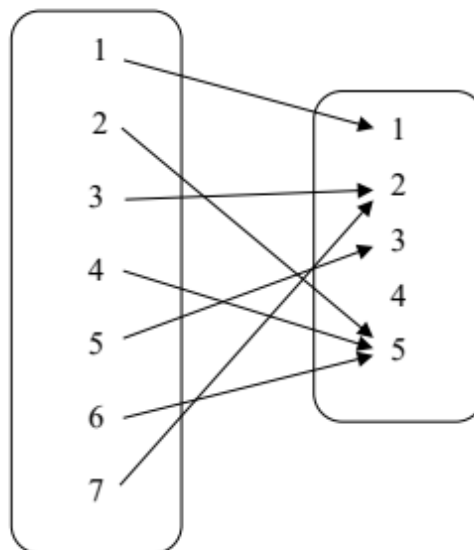
10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

- 1 What type of relation is shown below?



- A. One-to-One
- B. One-to-Many
- C. Many-to-One
- D. Many-to-Many
- 2 If $f(x) = x^2 + 1$ and $g(x) = \sqrt{3-x}$, then the domain for $f(x) \times g(x)$ is:
- A. $(-\infty, \infty)$
- B. $[3, \infty)$
- C. $(-\infty, 3]$
- D. $[-3, 3]$

3 What is the solution(s) of the equation $\log_2(x-3) + \log_2 x = 2$?

A. $x = -1, x = 4$

B. $x = -1$

C. $x = 4$

D. $x = 1, x = -4$

4 Consider the set of scores:

9 17 20 23 24 27 27 28 30 31 39 47

Which of the scores could be considered an outlier?

A. Only 9.

B. Only 47.

C. Both 9 and 47.

D. None of the scores.

5 What is the value of $\int_{-2}^2 |x-1| dx$?

A. 0

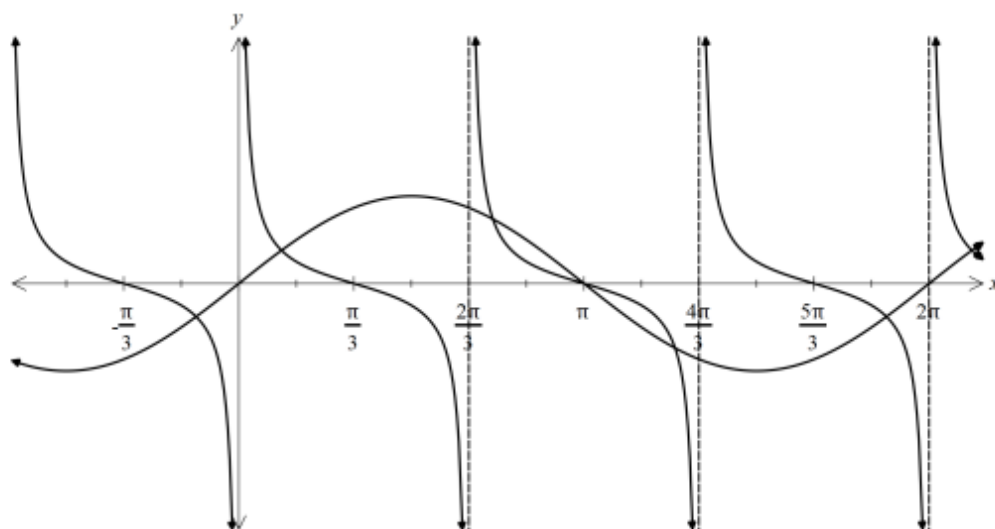
B. 2

C. 4

D. 5

- 6 For what values of k does the equation $4(k+1)x^2 + 4kx + (k-2) = 0$ have exactly two distinct roots?
- A. $k > -2$
B. $k < -2$
C. $k \geq -2$
D. $k \leq -2$
- 7 The curve $y = f(x)$ has a maximum turning point at $(2, 5)$.
For all values of x , $f(-x) = -f(x)$. Which of the following statements is true?
- A. $(-2, 5)$ is a minimum turning point of $y = f(x)$.
B. $(-2, -5)$ is a minimum turning point of $y = f(x)$.
C. $(-2, -5)$ is a maximum turning point of $y = f(x)$.
D. $(-2, 5)$ is a maximum turning point of $y = f(x)$.
- 8 Suppose that $\int_{-3}^8 f(x) dx = A$ and $\int_2^8 f(x) dx = B$.
The value of $\int_2^{-3} (f(x) + 2x) dx$ is:
- A. $A - B - 5$
B. $B - A + 5$
C. $A - B + 5$
D. $B - A - 5$

- 9 Below are the graphs of $y = 4 \sin(x)$ and $y = \cot\left(\frac{3x}{2}\right)$.



How many solutions are there of $4 \sin(x + \pi) - \cot\left(\frac{3x}{2}\right) = 0$ for $0 \leq x \leq 2\pi$?

- A. 2
 B. 3
 C. 4
 D. 5
- 10 What is the value of $\sum_{n=4}^n \ln 3^{2n}$?

- A. $2n \ln 3$
 B. $n(n+4) \ln 3$
 C. $(n-4)(n+4) \ln 3$
 D. $(n-3)(n+4) \ln 3$

End of Section I

Section II

90 marks

Attempt questions 11 to 34

Allow about 2 hours and 45 minutes for this section

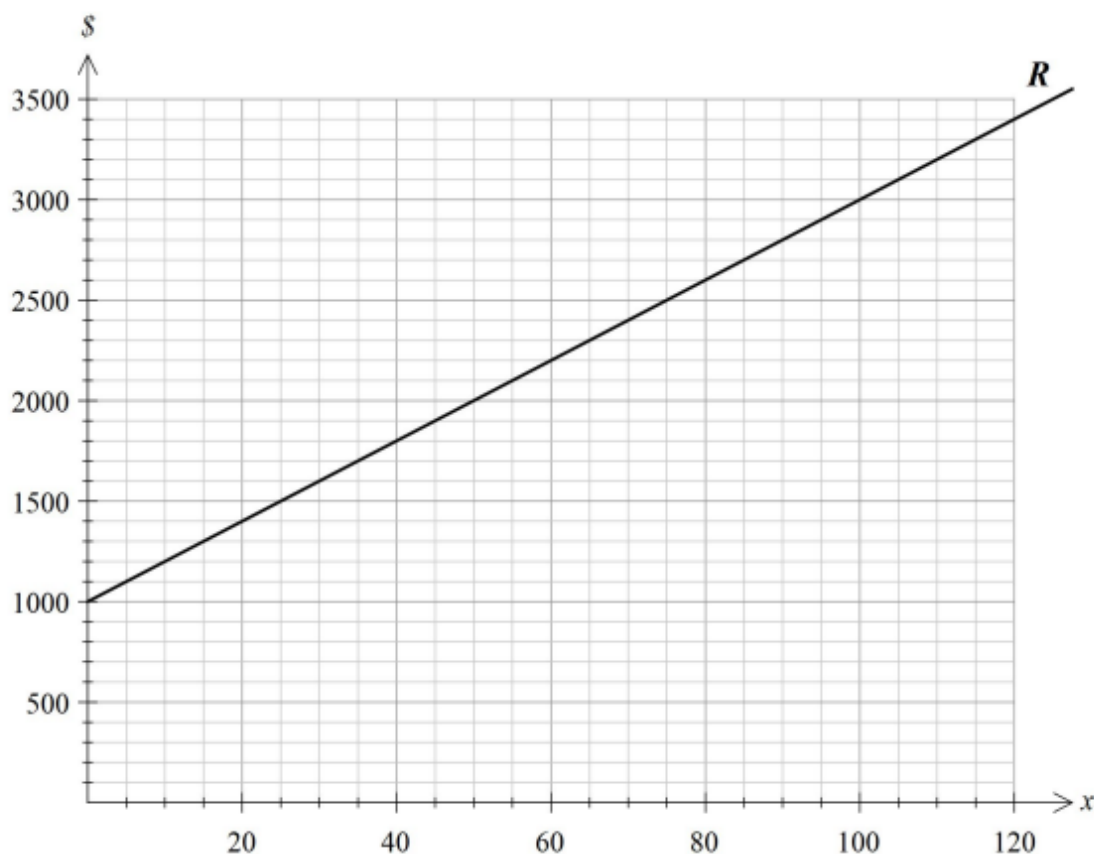
Answer each question in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of the examination paper.

Question 11 (4 marks)

The incoming revenue for a catering company, R , is based on a \$1 000 hiring fee plus \$20 per person where x is the number of guests. The graph of the revenue model has been provided on the grid below.



(a) Write the equation for R in terms of x .

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Question 11 continues on page 9

Question 11 (continued)

The cost for operating the catering company is modelled by $C = 1200 + 10x$, where C is the cost and x is the number of guests.

(b) On the grid on the previous page, draw the graph of this cost model and label it as C . **1**

(c) Using the graphs on the grid, or otherwise, find the value of x at the break-even point. **1**

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(d) Using the graphs, or otherwise, find the value of x when a profit of \$500 is made. **1**

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End of Question 11

Question 12 (3 marks)

A and B are two events such that $P(A) = 0.3$, $P(B) = 0.5$ and $P(A|B) = 0.4$.

(a) Show that $P(A \cup B) = 0.6$.

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(b) Find in simplest exact form $P(\text{neither } A \text{ nor } B)$.

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Question 13 (2 marks)

Evaluate $\int_{-1}^3 \frac{5x^3 - 6x^2 + 3}{x^2} dx$.

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Question 14 (4 marks)

The table shows the results of a survey of 300 people who have been vaccinated against a new virus.

(a) Complete the two-way table for the data collected.

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	Vaccinated	Not vaccinated	Totals
Infected	27		
Not infected		43	
Totals	137		300

(b) A person is selected at random from this group.

(i) Find the probability that the selected person is infected.

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(ii) Find the probability that the selected person is not vaccinated, given that they are infected.

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Question 15 (5 marks)

A car leaves a petrol station and drives $N 30^\circ W$ at 80 km/h. At the same time, a motor bike leaves the petrol station and travels due east at 60 km/h.

(a) Show that after one hour, the distance between the car and the motor bike is $20\sqrt{37}$ km.

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(b) Find the bearing of the car from the motor bike.
Give your answer correct to the nearest degree.

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Question 16 (4 marks)

Find the coordinates and nature of the stationary points on the curve $y = \frac{x^2}{x-2}$.

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[illegible]

Question 17 (2 marks)

Find $\int \frac{2x-1}{2x+1} dx$.

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Question 18 (3 marks)

(a) Differentiate $y = x \log_e x$.

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(b) Hence, find $\int (\log_e x + 1) dx$.

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Question 19 (2 marks)

Which term of the arithmetic sequence 3, 16, 29, ... is 224?

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Question 20 (2 marks)

The table below shows points on a continuous curve $y = f(x)$.

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x	3	3.2	3.4
$f(x)$	7.19	7.62	8.41

Use the trapezoidal rule to find the approximate value, correct to 2 decimal places, of

$$\int_3^{3.4} f(x) \, dx.$$

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Question 21 (6 marks)

Trout are released into a lake for the first time. The size of the population, P , of the trout can be modelled by the function $P = \frac{A}{1 + 15e^{-0.5t}}$, where t is the time in months after the trout are introduced into the lake.

- (a) Initially, 250 trout were released into the pond. Show that $A = 4\,000$. **1**

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- (b) Find the range of the function P . **1**

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- (c) At what rate is the population growing two months after the trout were released into the Lake? Give your answer correct to the nearest number of trout. **2**

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Question 21 (continued)

(d) How many months will it take for the population of trout to exceed 3 000?

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Give your answer as a whole number.

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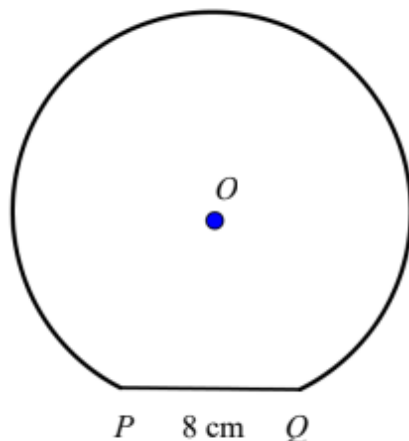
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End of Question 21

Question 22 (4 marks)

A table tennis bat (without the handle attached) is in the shape of a major segment of a circle, with centre O and radius 8 cm. The length of the straight edge PQ is also 8 cm.

NOT TO SCALE



- (a) Explain why $\angle POQ = \frac{\pi}{3}$.

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- (b) Find the exact area of the table tennis bat.

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Question 23 (2 marks)

Prove that $(1 + \sec x + \tan x)(1 - \sec x + \tan x) = 2 \tan x$.

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Question 24 (3 marks)

(a) Find the limiting sum of the geometric series $3 + \frac{3}{\sqrt{3}+1} + \frac{3}{(\sqrt{3}+1)^2} + \dots$.

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(b) Explain why the geometric series $3 + \frac{3}{\sqrt{3}-1} + \frac{3}{(\sqrt{3}-1)^2} + \dots$ does not have a limiting sum.

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Question 25 (5 marks)

A discrete random variable X with mean 4 has the probability distribution below for some constants a and p .

x	a	$2a$	$3a$	$4a$
$p(x)$	$4p$	$3p$	$2p$	p

(a) Find the values of a and p .

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(b) Find $\text{Var}(X)$.

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(c) Find $P(X \leq 6 | X > 3)$.

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Question 26 (4 marks)

The circle $x^2 + ax + y^2 + by + c = 0$ with a radius of 3 units, is shifted to the right by 1 unit.

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After the shifting, the circle has the centre at $(3,4)$. Find the value of a , b and c .

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Question 27 (3 marks)

Evaluate the sum of the geometric series $4+12+36+ \dots +26\,244$.

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Question 28 (6 marks)

A truck travels between two towns. The distance between the towns is B km.

The trucker has the following costs:

The hourly fuel costs are proportional to the square of the speed.

The hourly fuel cost is \$2000 when the speed is 40 km/h.

All other costs are \$4800 per hour. Let x be the speed in km/h and C be the total cost.

(a) Show that $C = B\left(\frac{5}{4}x + \frac{4800}{x}\right)$.

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Question 28 continues on page 23

Question 28 (continued)

(b) What is the most economical speed?

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[illegible]

End of Question 28

Question 29 (4 marks)

A teacher is interested in the relationship between how much time their class of 10 students spends on their phone with that of their laptop.

The teacher collected the results over the month of June and the average number of hours spent per day for each student is outlined in the table below.

STUDENT	A	B	C	D	E	F	G	H	I	J
Hours on phone per day (x)	7.6	7.0	8.9	3.0	3.0	7.5	2.1	1.3	5.8	6.2
Hours on laptop per day (y)	1.7	1.1	0.7	5.8	5.2	1.7	6.9	7.1	3.3	4.4

- (a) Find the equation of the Least-Squares Regression Line.

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Give each coefficient to 3 decimal places.

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- (b) Calculate Pearson's Correlation Coefficient, correct to 3 decimal places.

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What conclusion can be made about the time students spend on their phone compared to their laptops?

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Question 30 (6 marks)

A moving particle has an acceleration of $a \text{ m/s}^2$ at time t seconds ($t \geq 0$).

The acceleration is given by the equation

$$a = 15 - 6t.$$

Initially the particle is at the origin and has a velocity of $v = -12 \text{ m/s}$.

- (a) Find the velocity, v , and displacement, x , in terms of t . **2**

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- (b) Find the time(s) when the particle is at rest. **1**

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- (c) Find the distance travelled during the first 4 seconds. **3**

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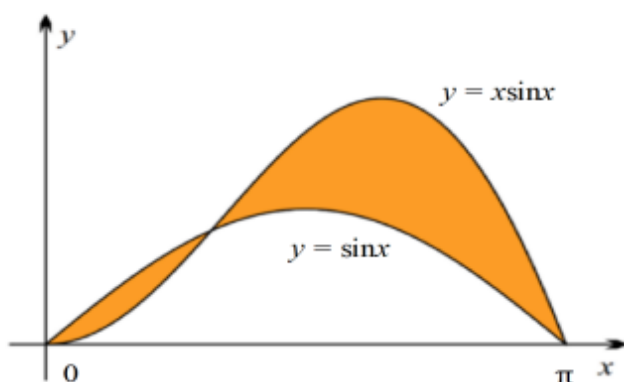
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Question 31 (6 marks)

A sailing club has designed a new logo which is the shaded regions shown in the diagram below.

The shaded regions are bounded by $y = \sin x$ and $y = x \sin x$ in the interval $0 \leq x \leq \pi$.



NOT TO SCALE

- (a) Find $\frac{d}{dx}(x \cos x)$ and hence show $\int x \sin x \, dx = \sin x - x \cos x + C$.

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- (b) Find the area of the logo. Give your answer in exact form.

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Question 32 (3 marks)

Let $f(x) = e^{-kx} + 3x$, where k is a positive real number.

- (a) Find, in terms of k , the x -coordinate of the stationary point of the graph of $y = f(x)$. **2**

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- (b) State the values of k such that the x -coordinate of this stationary point is a positive real number. **1**

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Question 33 (3 marks)

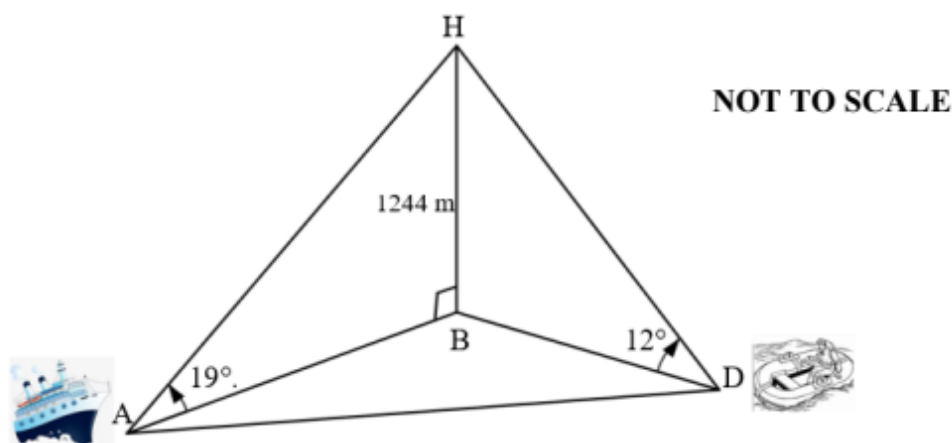
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David is in a life raft, D, and Anna is in a cabin cruiser, A, searching for him.

They are in contact by mobile telephone. David tells Anna that he can see the top of the mountain Mt Hope, H.

From David's position, the mountain has a bearing of 340° and the angle of elevation to the top of the mountain is 12° .

Anna can also see Mt Hope. From her position, the mountain has a bearing of 070° and the top of the mountain has an angle of elevation of 19° . The top of Mt Hope is 1 244 m above sea level.



Find the distance of the life raft from Anna's position.

Give your answer correct to two decimal places.

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Question 34 (4 marks)

On a particular day, the depth, D metres, of water in Blue Harbour is given by the function $D = 8 + 2 \cos\left(\frac{4\pi}{25}t\right)$ for $0 \leq t \leq 24$, where t is the number of hours after midnight.

(a) Find the amplitude and period of the function. **2**

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(b) Find the first time when the water depth is 7 metres. **2**

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End of paper